

Add-On Lab: Give Your MCP Server a Memory

This is a small addition to the MCP server you already built, same repo, same company, same problem. Pick ONE of the two options below and build it.

Option A: Add Search Over Your Data (RAG)

Give your server one new tool that can search a piece of unstructured content in your domain: a document, notes, reports, instead of the model reading it all at once.

- Pick one document-shaped data source in your domain (e.g., past visit notes, policy text, case history).
- Chunk it, and store it (in memory is fine)
- Add one new tool: `search_knowledge_base(query, top_k)` with a typed schema, same style as your other tools.
- Show one demo query that only this tool could answer correctly.

Option B: Give Your Agent Memory

Let your agent remember one thing about a specific entity (a client, a pet, a ticket) between two separate sessions.

- Pick the entity your memory is scoped to.
 - When something worth remembering happens, write a short structured record (what happened, when, outcome) to a vector store, tagged with that entity's ID.
 - At the start of a new session, look up and load the most relevant past records for that entity.
 - Demo it in two runs: session 1 where a fact comes up, session 2 (separate run) where the agent recalls it on its own.
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What to Turn In

- A new folder in your repo (`mcp_server/rag/` or `mcp_server/memory/`) with your code.
- One GitHub Issue for this work, with a real reason it needed to happen not just a task title.
- A short demo (a few messages is enough) showing it working.